

MODULE 20

# ER Diagrams and Schema Design

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DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

# ER Diagrams and Schema Design

## ER Diagram Components

Entity: real-world object (Students, Courses)

Attribute: property of entity (name, age, gpa)

Relationship: how entities connect (enrolls in)

Cardinality: how many (1:1, 1:N, M:N)

## Cardinality Types

One-to-One 1:1 Person --- Passport

One-to-Many 1:N Department --- Students  
(one dept has many students)

Many-to-Many M:N Students --- Courses  
(requires a junction table!)

## University Schema

departments(dept\_id PK, name, head\_id)

instructors(inst\_id PK, name, dept\_id FK)

courses(course\_id PK, title, credits, inst\_id FK)

students(student\_id PK, name, dept\_id FK)

enrollments(student\_id FK, course\_id FK, grade, semester)

## Design Best Practices

Every table needs a PRIMARY KEY

Use FOREIGN KEYS for relationships

Plural table names: students, not student

Consistent snake\_case naming

Add created\_at / updated\_at timestamps

Normalize to at least 3NF

## Blog Schema Example

```
CREATE TABLE users (  
  user_id INT AUTO_INCREMENT PRIMARY KEY,  
  username VARCHAR(50) UNIQUE NOT NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
CREATE TABLE posts (  
  post_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
user_id INT NOT NULL,  
title VARCHAR(200),  
FOREIGN KEY(user_id) REFERENCES users(user_id)  
ON DELETE CASCADE  
);
```